# MILITARY STANDARD

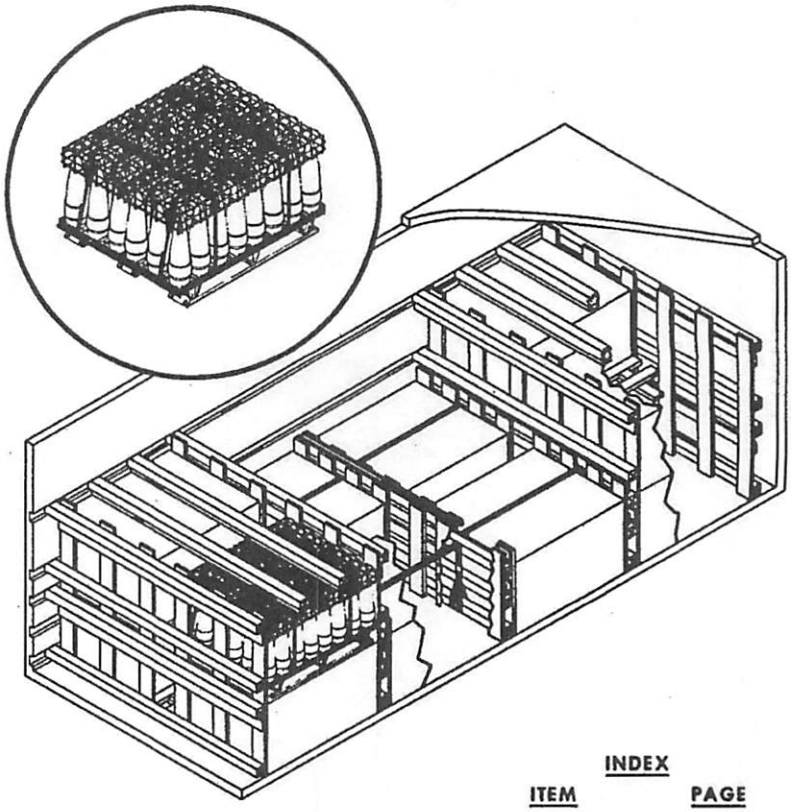
**MIL-STD-1386-25**
**(NAVY)**
10 AUGUST 1973

## LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

### 5"/38 PROJECTILE
### FLEET ISSUE UNIT LOAD

**UNIT LOAD DATA**

UNIT LOAD DWG ..... WR-54/4
DOT HAZARD CLASSIFICATION .... EXPLOSIVES A
COAST GUARD CLASS ..... V
OTHER ..... SEE TABLE I.



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236(NAVY).

**INDEX**


| ITEM | PAGE |
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| COFC & CONTAINERSHIP | 2, 3, & 5 |


FSC 8140

**AUTHORIZED FOR COFC, CONTAINERSHIP AND MOVEMENT ABOARD DEPOT (NON-PUBLIC) ROADS ONLY.**

**APPROVED BY US COAST GUARD**
[illegible]
DATE 6-19-73

**APPROVED BY BUREAU OF EXPLOSIVES**
A. F. Grasmick
DATE 6/19/73

SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

SIGNATURE (ORD) SYSCOM, BY DIRECTION DATE

**ORIGINATOR** SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY



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# GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOADS OF 5"/38 PROJECTILES. THE PROJECTILES SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/4.
2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.
3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR.

**CAUTION**

WHEN POSITIONING CROSSMEMBERS, MAKE A VISUAL CHECK TO ENSURE THAT THE CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

4. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO BELT RAILS FOR SHIPMENT.
5. WHEN THE VOID BETWEEN THE CROSSMEMBER AND THE SEPARATOR GATE IS GREATER THAN ONE INCH, USE FILL MATERIAL TO SHIM BETWEEN THE CROSSMEMBER AND GATE AS SHOWN IN DETAIL G, PAGE 4.
6. SWAY BRACE, DETAIL A, IS ASSEMBLED IN PLACE. PIECE 1 IS POSITIONED INBOARD OF THE CORNER PALLET POSTS. WITH PIECES 2 AGAINST PALLET SKIDS, NAIL TO PIECE 1 WITH THREE 10d NAILS PER JOINT. CUT PIECE 3 TO FIT BETWEEN PIECES 2 AND NAIL TO PIECE 1 WITH THREE 10d NAILS.
7. HEIGHT OF HOLD DOWN ASSEMBLY, DETAIL J, MAY BE VARIED TO ACCOMODATE DIFFERENCES IN UNIT LOAD HEIGHTS BY ADJUSTING PIECE SIZE OR OMITTING PIECES AS REQUIRED. STRUCTURAL INTEGRITY OF HOLD DOWN ASSEMBLY SHALL BE MAINTAINED. VOID BETWEEN HOLD DOWN CROSSMEMBER AND PIECE NO. 3 OF HOLD DOWN ASSEMBLY SHALL NOT EXCEED ONE HALF INCH.
8. SHIM, PIECE NO. 10, SHALL BE OF SUFFICIENT THICKNESS TO FILL THE VOID BETWEEN THE TOP OF THE UNIT LOAD AND THE BOTTOM OF THE CROSSMEMBER. ONE HALF INCH SLACK IS PERMITTED.
9. POSITION SEPARATOR GATES, DETAILS C AND D WITH UPRIGHTS, PIECES 4 AND 6, AGAINST UNIT LOAD AND HORIZONTALS, PIECES 5 AND 7, AWAY FROM UNIT LOAD.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC. MM-L-751
    NAILS - FED. SPEC. FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT
    STRAP - FED. SPEC. QQ-S-781, TYPE I, CLASS A
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOLUME 1).

**TABLE I**

| ITEM | DIMENSIONS | CUBE (CU FT) | UNIT LOAD WT. (LBS) PALLET ADAPTER MK 11 | | CONTAINER NET WT. (LBS) PALLET ADAPTER MK 11 | | CONTAINER GR.WT. (LBS) PALLET ADAPTER MK 11 | |
|---|---|---|---|---|---|---|---|---|
| | | | MOD 0 | MOD 1 | MOD 0 | MOD 1 | MOD 0 | MOD 1 |
| PROJECTILE WITHOUT PROTECTING CAP | 48 L x 40 W x 27 1/2 H | 30.5 | 2965 | 2910 | 35,580 | 34,920 | 41,830 | 41,170 |
| PROJECTILE WITH WINDSHIELD | 48 L x 40 W x 27 1/2 H | 30.5 | 2965 | 2910 | 35,580 | 34,920 | 41,830 | 41,170 |
| PROJECTILE WITH PROTECTING CAP | 48 L x 40 W x 28 3/8 H | 31.6 | 3085 | 3030 | 37,020 | 36,360 | 43,377 | 42,717 |

# COFC AND CONTAINERSHIP ONLY

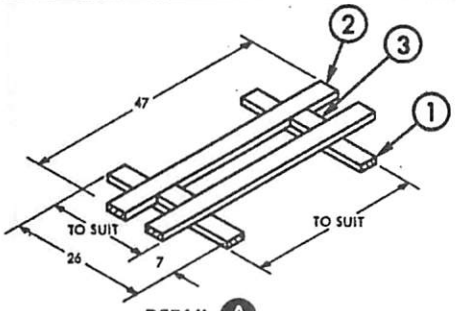
1. THE CONTAINER LOAD CONSISTS OF 12 UNIT LOADS WHICH SHALL BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT.
2. WHEN REQUIRED TO LOAD UNIT LOADS OF:
   (a) PROJECTILES WITHOUT PROTECTING CAPS, OR
   (b) PROJECTILES WITH WINDSHIELDS
   FOLLOW THE REQUIREMENTS SHOWN ON PAGE 3.
3. WHEN REQUIRED TO LOAD UNIT LOADS OF PROJECTILES WITH PROTECTING CAPS, FOLLOW PROCEDURES OF PAGE 3 WITH THE FOLLOWING EXCEPTION:
   (a) THE TWO HIGH STACKED UNIT LOADS (THE 1ST AND LAST BAYS) SHALL BE TOMMED DOWN AS SHOWN ON PAGE 5.

**WARNING**

BECAUSE THE WEIGHT LOADED INTO THIS CONTAINER EXCEEDS 34,000 LBS, THE MILVAN CHASSIS CANNOT BE USED TO TRANSPORT THE LOADED CONTAINER ON PUBLIC HIGHWAYS. THE LOADED CONTAINER MAY BE TRANSPORTED OVER DEPOT ROADS USING MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY.



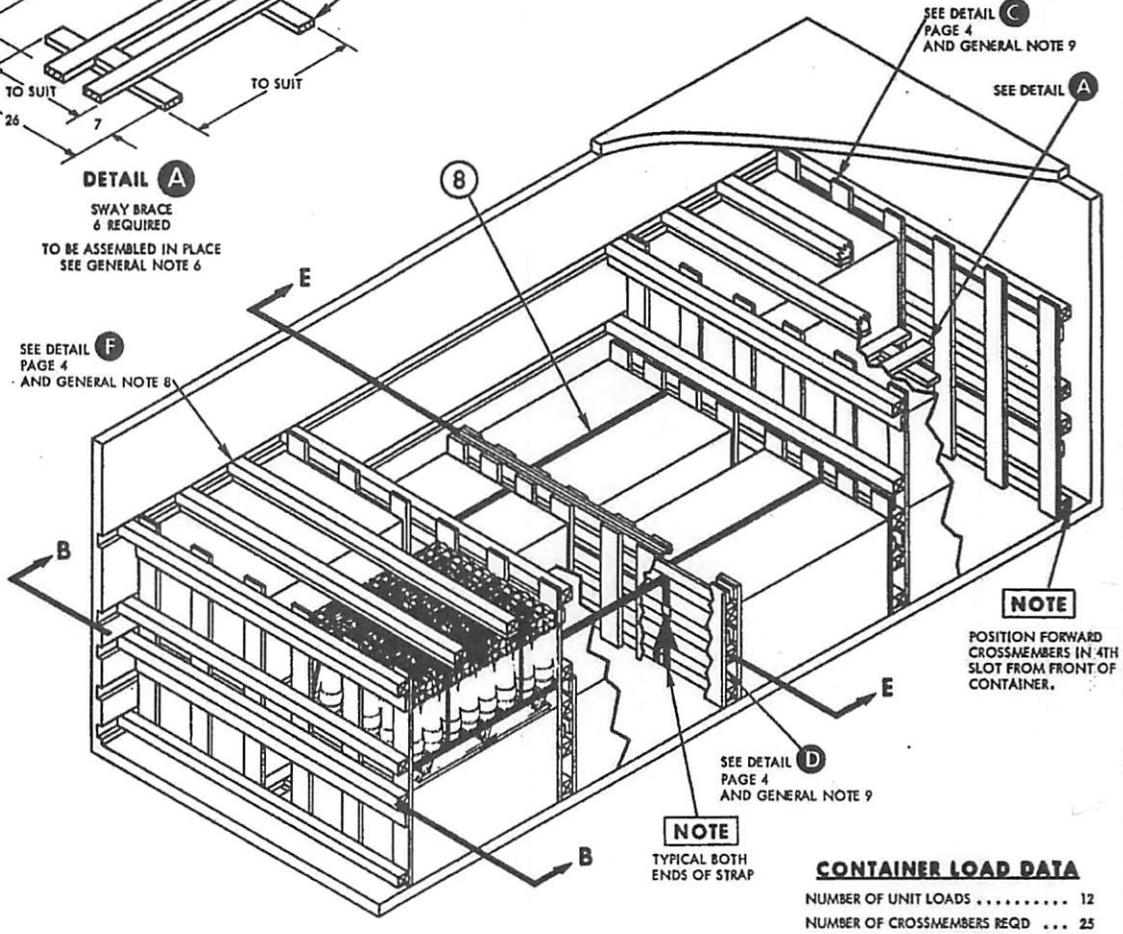
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**DETAIL A**

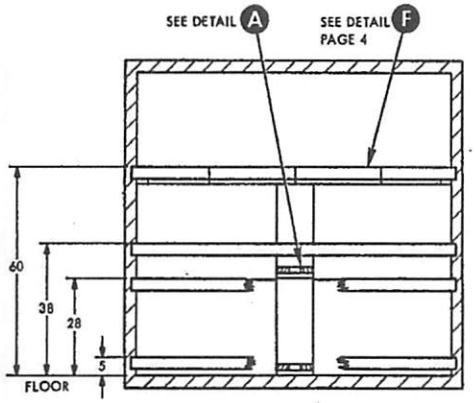
SWAY BRACE
6 REQUIRED

TO BE ASSEMBLED IN PLACE
SEE GENERAL NOTE 6




## CONTAINER LOAD DATA

NUMBER OF UNIT LOADS ..... 12
NUMBER OF CROSSMEMBERS REQD ... 25
CONTAINER NET WEIGHT .... 5785 LBS
DUNNAGE WEIGHT ..... 465 LBS
OTHER DATA ..... SEE TABLE I




**SECTION B-B**

SEPARATOR GATE, DETAIL C
OMITTED FOR CLARITY

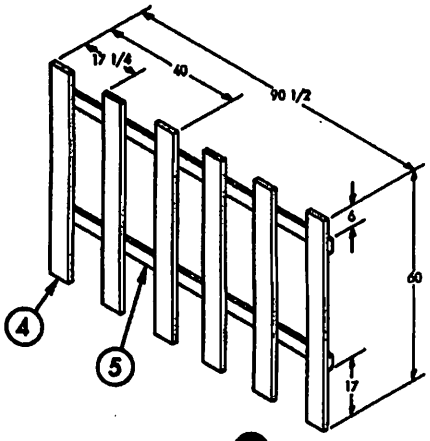
| PIECE NO. | DESCRIPTION | SIZE | NO. PCS REQD | NAIL TO | NAILS NUMBER | NAILS SIZE |
|---|---|---|---|---|---|---|
| 10 | SHIM | THICKNESS TO SUIT x 4 x 90 | 4 | - | - | - |
| 9 | SEAL | FOR 1 1/4 STRAP | 8 | - | - | - |
| 8 | STRAP | 1 1/4 x .035 x 16 FT | 4 | - | - | - |
| 7 | HORIZONTAL | 1 x 4 x 90 1/2 | 8 | SEE 6 | - | - |
| 6 | UPRIGHT | 1 x 6 x 33 | 32 | 7 | 3 PER JOINT CLINCHED | 8d |
| 5 | HORIZONTAL | 1 x 4 x 90 1/2 | 8 | SEE 4 | - | - |
| 4 | UPRIGHT | 1 x 6 x 60 | 24 | 5 | 3 PER JOINT CLINCHED | 8d |
| 3 | SPACER | 2 x 4 TO SUIT | 12 | 1 | 3 PER JOINT | 10d |
| 2 | STRINGER | 2 x 4 x 47 | 12 | 1 | 3 PER JOINT | 10d |
| 1 | SUPPORT | 2 x 4 x 26 | 12 | SEE 2 & 3 | - | - |

LIST OF MATERIALS AND NAILING DATA

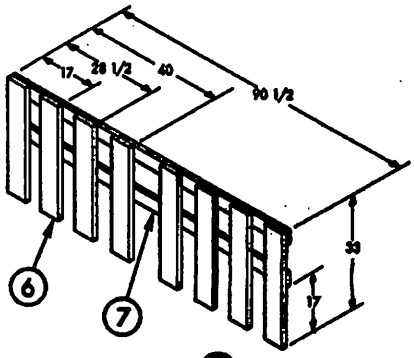
**PROJECTILE W/O PROTECTING CAP**
**PROJECTILE WITH WINDSHIELD**



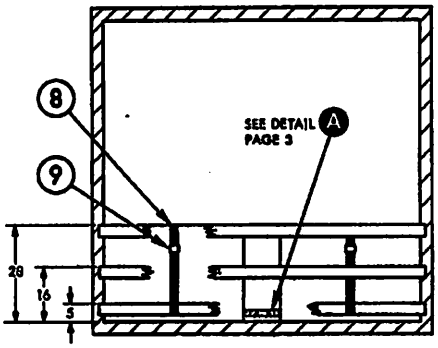
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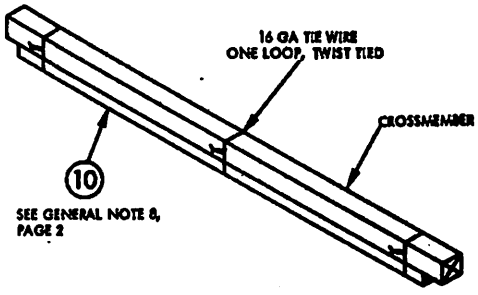

DETAIL C
SEPARATOR GATE
4 REQUIRED
(SEE GENERAL NOTE 9, PAGE 2)



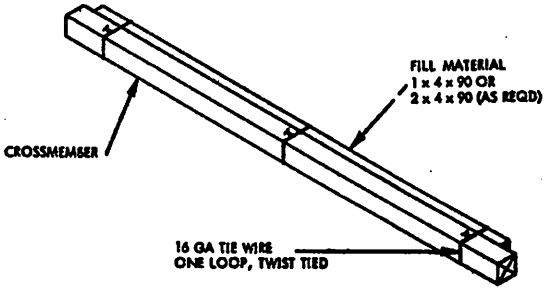

DETAIL D
SEPARATOR GATE
4 REQUIRED
(SEE GENERAL NOTE 9, PAGE 2)




SECTION E-E
SEPARATOR GATE, DETAIL D
OMITTED FOR CLARITY



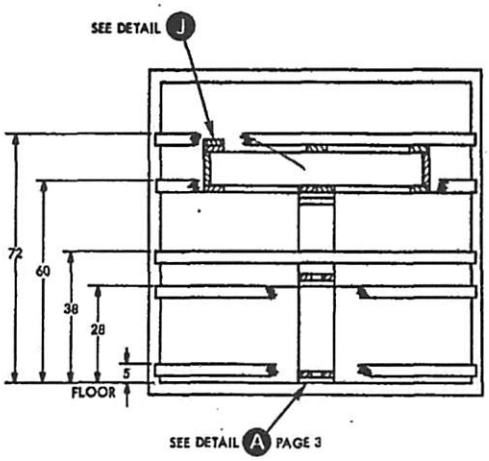

DETAIL F
HOLD DOWN
4 REQUIRED




DETAIL G
CROSSMEMBER WITH
FILL MATERIAL
SEE GENERAL NOTE 8, PAGE 2



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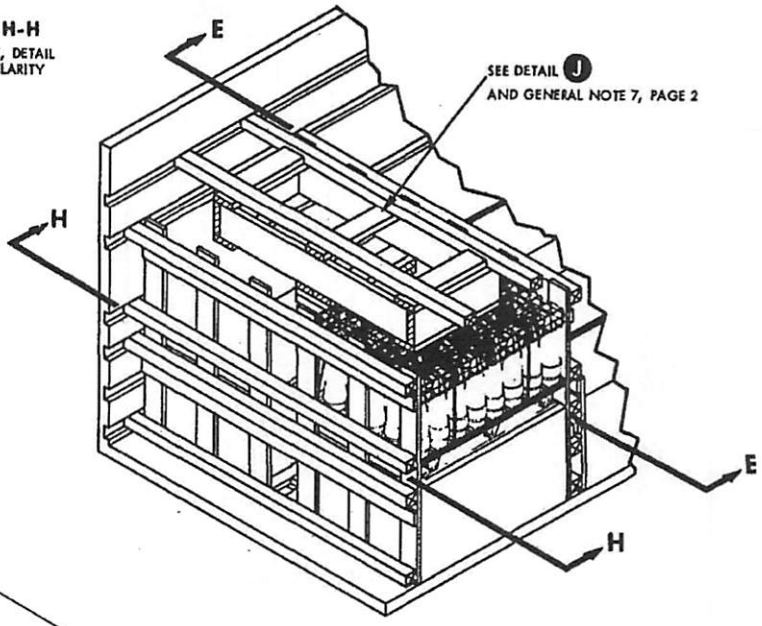


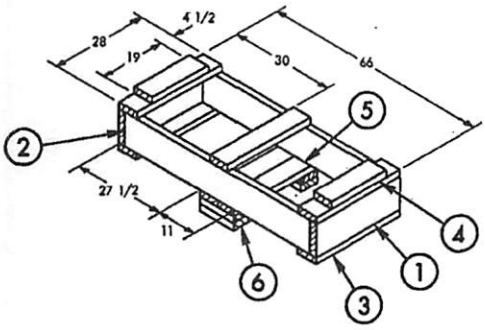

## CONTAINER LOAD DATA

NUMBER OF UNIT LOADS ..... 12
NUMBER OF CROSSMEMBERS REQD ... 25
CONTAINER NET WEIGHT .... 5785 LBS
DUNNAGE WEIGHT ..... 572 LBS
OTHER DATA ..... SEE TABLE I, PAGE 2

**SECTION H-H**
SEPARATOR GATE, DETAIL OMITTED FOR CLARITY

**NOTE**
CENTER HOLD-DOWN DETAIL J LONGITUDINALLY IN BAY.




**DETAIL J**
HOLD DOWN
2 REQUIRED

| PIECE NO. | DESCRIPTION | SIZE | NO. PCS REQD | NAIL TO | NAILS NUMBER | NAILS SIZE |
|---|---|---|---|---|---|---|
| 6 | SPACER | 2 x 6 x 11 | 4 | 6<br>5 | 5<br>4 PER JOINT | 12d<br>12d |
| 5 | SUPPORT | 2 x 6 x 28 | 4 | 2 | 4 PER JOINT | 12d |
| 4 | STOP | 2 x 6 x 19 | 4 | 3 | 4 PER JOINT | 12d |
| 3 | SHIM | 2 x 6 x 28 | 10 | 1 & 2 | 3 PER JOINT | 16d |
| 2 | CROSSMEMBER | 2 x 10 x 63 | 4 | SEE 1 | - | - |
| 1 | ENDMEMBER | 2 x 10 x 28 | 4 | 2 | 4 PER JOINT | 16d |

LIST OF MATERIALS AND NAILING DATA



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REVIEW ACTIVITY:
NAVY-OS

PREPARING ACTIVITY:
NAVY-OS
(PROJECT NO. 6140-N206)



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